THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A device connected to a 5 A fuse draws 3 A and a second device draws 4 A. If both are on in the same circuit, the fuse will: 1

(a) cool **(b)** become magnetic **(c)** melt **(d)** remain intact

2. The same heater is run for 2 hours instead of 1 hour. The heat produced is: 1

(a) zero **(b)** halved **(c)** doubled **(d)** same

3. Assertion (A): The bulb filament is made of tungsten. Reason (R): Tungsten has a very low melting point. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A long straight wire runs north to south just above a compass needle that points north-south. When a current is switched on in the wire, the needle: 1

(a) melts **(b)** stays and points more strongly north
(c) deflects from its north-south position **(d)** does not move

5. An iron's flex has a thin insulation burnt off. A user touching it would risk: 1

(a) an electric shock **(b)** a bulb glow
(c) a cold **(d)** a magnet

6. Assertion (A): Household appliances are connected in parallel. Reason (R): Each appliance receives the same voltage and works independently. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Why can a bird sitting on a single bare overhead power line be safe? 1

(a) wires are cold
(b) no current passes through the bird since there is no potential difference across its feet
(c) birds are insulators
(d) feathers are magnets

8. A fuse blows frequently in a house. The best action is to: 1

(a) avoid switches
(b) put a nail in the fuse
(c) use a thicker wire as fuse
(d) find and remove the cause such as overloading

9. **Assertion (A):** A fuse wire is made of a material with a high melting point. 1
Reason (R): A fuse protects a circuit by melting and breaking it when the current becomes too large.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The filament of an electric bulb glows when current passes through it. 1
Reason (R): A bulb filament is strongly cooled by its surroundings so it never gets hot.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A student uses a nail without winding wire and connects it to a cell. Will it become a magnet? Justify. 2

12. Why is the heating coil in an iron wound in a coil and not stretched out? 2

13. Why does a fuse protect even though it is the weakest link in the circuit? 2

14. Why is heating of a wire a problem in transmission lines? Name one remedy. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how a short circuit leads to fire. How does a fuse or MCB prevent it? 3

16. Explain why electric current produces both heat and a magnetic field at the same time in a wire. 3

17. A thin wire becomes red hot and a thick one stays cool. Use this to explain why heating elements are thin. 3

OR

A student's electromagnet lifts 5 pins with a new cell but only 2 pins with an old one. Explain and give a conclusion. 3

18. Explain how the strength of an electromagnet can be tested quantitatively. 3

OR

Explain the function of each part of a simple electric bell: coil, soft iron core, armature, hammer, gong. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a house wiring, the normal current drawn by all appliances on a circuit is 8 A. The electrician has fuse wires rated 5 A, 10 A and 30 A. During a short circuit the current rises suddenly and the lights go out, though the wiring is undamaged. 5
(i) What is a fuse?
(ii) Which fuse should be used for this circuit?
(iii) Which effect of current makes a fuse work?
(iv) Why should a 30 A fuse not be used here?

20. A student made a circuit with a bulb and found that touching the wire for a long time made it hot. Explain the heat source, why it is higher in a thin wire and why it matters for wiring in houses. 5

OR

A wire A (nichrome, thin) and a wire B (copper, thick) are connected in series with a cell. Predict which gets hotter. Then explain what happens if they are connected in parallel. 5

END OF QUESTION PAPER — **Total: 40 Marks**